

	minimum.	
5(a)	The root-mean-square (r.m.s.) voltage of an alternating current is defined as the <u>steady direct voltage</u> which <u>converts electrical energy to other forms of energy in a given resistance at the same rate</u> as the a.c.	
5(b)(i)	Turns Ratio $\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{240 \times 10^3}{12.0 \times 10^3}$ $= 20$	
5(b)(ii)	Transmitting current = $\frac{\text{power}}{\text{voltage}} = \frac{20 \times 10^6}{240 \times 10^3}$ $= 83.3 \text{ A}$ Power lost = $I^2 R = (83.3)^2 (200) = 1387778 \text{ W}$ $= 1390 \text{ kW}$	
5(b)(iii)	1 day 24 hrs Power lost in kWh = $24 \times 1387.78 = 33360 \text{ kWh}$ Amount lost = $33360 \times 0.1 = \$3340$	

6(a)	When $V = 1.4 \text{ V}$, $I = 2.5 \text{ mA}$	
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